

Course Code : ECT 353

KQLR/RS – 24 / 1606

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

MICROCONTROLLERS AND INTERFACING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Due credit will be given to neatness.

1. (a) Answer the following questions :
 - (i) Recall the functions of low level active PSEN and EA signals ?
 - (ii) Compare polling with interrupt method with respect to efficient handling of activities. 4(CO1)
- (b) Sketch the interface of a dual DAC to the 8051 microcontroller. The DAC output is connected to an analog comparator where the reference of comparison is 3 V. A LED is connected to the output of the comparator for indication. Write an 8051 assembly program segment to flash the LED only when the output value given to DAC greater than the comparison. 6(CO3,5)
2. (a) Construct and explain an interfacing model of 8051 microcontroller such that external program memory of 8 KB and data memory of 8 KB is interfaced. 5(CO3)
- (b) Write an assembly language program using 8051 to transfer an array of 10 bytes stored in locations starting from 8000H in 16K×8 Data RAM to locations starting from 9000H. Also draw the flow chart. 5(CO2,4)
3. (a) Write a program to perform the following :
 - (1) Keep monitoring P1.2 until it becomes high.
 - (2) When P1.2 becomes high write value 45H on P0.3.Sent a high to low pulse to P2.3. 4(CO2)

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- (b) How to interface Two 7 segment display interfacing with ARM7 across PORT 0 as per lower byte to higher Byte ? Write a program to display 00 to FF in incrementing pattern. 6(CO4,5)
4. (a) What are the different features of Thumb instruction set that make it suitable for embedded applications ? 5(CO1,5)
- (b) With a neat diagram explain the Addressing modes of ARM processors. 5(CO1)
5. (a) Discuss the interrupts or exceptions. How are they handled in ARM processors ? 5(CO1,4)
- (b) Illustrate the fundamental RISC and CISC design principles. 5(CO4,5)
6. (a) Examine elementary real-time task scheduling algorithms. 5(CO3,4)
- (b) How do hard real-time systems and soft real-time systems differ from one another ? 5(CO3)

